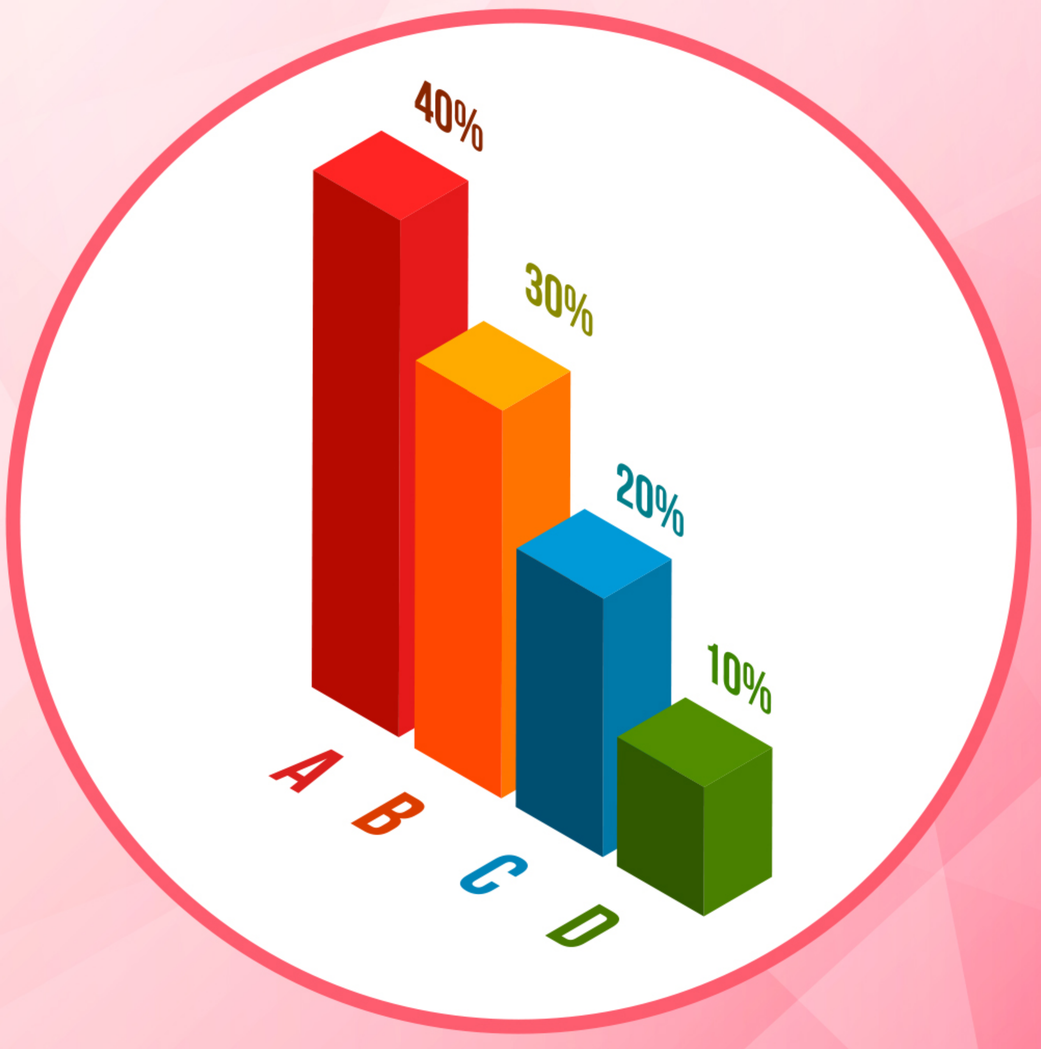
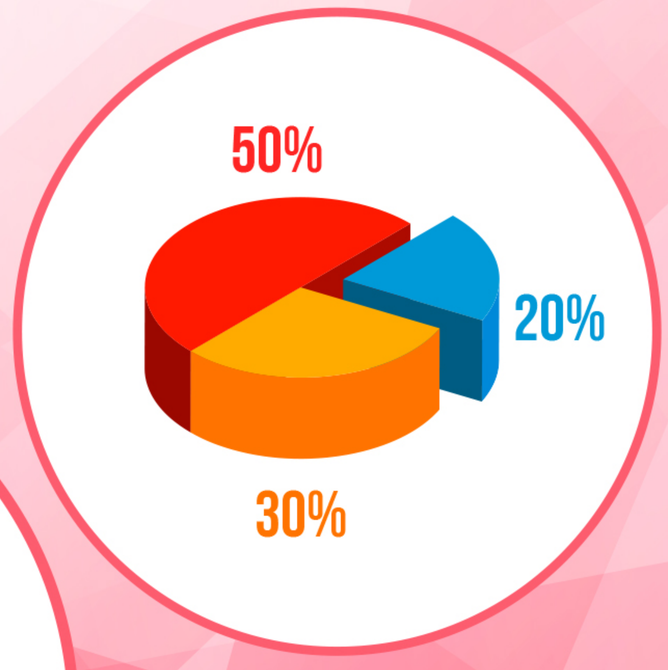
DATA

Tables & other important Concepts for DI

Assignment 02





hitbullseye.com

DI- Tables Online Assignment -02

DIRECTIONS ***for questions 1 – 5:*** Refer to the following table about how yearly periodic investments of equal amounts grow to Rs. 1,000 and answer the questions that follow.

	Amount Invested (Rupees) at an Interest Rate of:									
Year	**5%**	**6%**	**7%**	**8%**	**9%**	**10%**	**11%**	**12%**	**13%**	**14%**
1	952.38	943.40	934.58	925.93	917.43	909.09	900.90	892.86	884.96	887.19
2	464.58	457.96	451.49	445.16	438.96	432.90	426.97	421.16	415.47	409.90
3	302.10	296.33	290.70	285.22	279.87	274.65	269.56	264.80	259.75	255.03
4	220.96	215.65	210.49	205.48	200.61	195.88	191.26	186.82	182.47	178.25
5	172.36	167.36	162.51	157.83	153.30	148.91	144.66	140.54	136.56	132.70
6	140.02	135.25	130.65	126.22	121.94	117.82	113.85	110.02	106.33	102.77
7	116.97	112.39	107.99	103.77	99.72	95.82	92.09	88.50	85.05	81.75
8	99.74	95.32	91.09	87.05	83.19	79.49	75.96	72.59	69.37	66.29
9	86.37	82.10	78.02	74.15	70.46	66.95	6.3.61	60.43	57.41	54.53
10	75.72	71.57	67.64	63.92	60.39	57.04	53.88	50.88	48.04	45.36
11	67.04	63.01	59.21	55.63	52.24	49.06	46.05	43.23	40.57	38.07
12	59.83	55.92	52.24	48.79	45.55	42.51	39.66	37.00	34.50	32.17
13	53.77	49.96	46.40	43.08	39.97	37.07	34.37	31.85	29.51	27.34
14	48.59	44.89	41.44	38.24	35.26	32.50	29.94	27.56	25.37	23.34
15	44.14	40.53	37.19	34.10	31.25	28.61	26.18	23.95	21.90	20.01
16	40.26	36.75	33.51	30.53	27.80	25.29	22.99	20.88	18.96	17.21
17	36.86	33.44	30.30	27.43	24.81	22.42	20.24	18.26	16.47	14.84
18	33.85	30.53	27.49	24.72	22.21	19.94	17.88	16.02	14.34	12.83
19	31.19	27.94	25.00	22.34	19.94	17.77	15.82	14.07	12.51	11.11
20	28.80	25.65	22.80	20.23	17.93	15.87	14.03	12.39	10.93	9.64

1. To have periodic investments grow to Rs. 1,000 in 7 years, how much more will need to be invested annually at the 7% rate than at the 8% rate?

1. Rs. 4.05 2. Rs. 4.09 3.Rs. 4.22 4. Rs. 5.00

2. How many years will it take for annual investment of Rs. 25.65 invested at 6% to grow to Rs. 1,000?

1. 12 2. 14 3.16 4. 20

3. At a rate of 14%, how much more is the annual investment for a 10-year period than for a 14-year period?

1. Rs. 22.02 2. Rs. 23 3.Rs. 23.34 4. Rs. 45.36

4. What is the total amount that must be invested at 9% over a period of 4 years to have Rs. 1,000 at the end of this period?

1. Rs. 200 2. Rs. 352 3.Rs. 802 4. Rs. 502

5. For how many years and at which rate should periodic payments of less than Rs. 20 a year be invested in order to grow to Rs. 1,000?

1. 18 years at 8% 2.18 years at 9% 3.18 years at 10% 4. 20 years at 8%

DIRECTIONS*for questions 6 to 10:* *The financial profile of a company is as given below:*

(Rs million)	***Year ended 31st Mar 2001***	***Year ended 31st Mar 2000***
Gross Revenue	187.1	147.8
Gross Profit	10.8	3.9
Profit after Tax	8.8	3.1
Equity Capital	7.5	7.5
Reserves	18.4	20.0
EPS	117.33	41.33
Net Margin %	4.65	1.82

6. What is the % growth shown of gross revenue in 2001 over 2000? *(Figures shown in parenthesis represent negative growth)*

1. 18% 2. 27% 3.22% 4. 10%

7. The net margin in 2001 has shown an increase over 2000 of (Net margin is % of Gross revenue):

1. 3 m 2.6 m 3.0.6m 4. 1.2m

8. By how many percentage points has Gross Margin (Gross Profit as a % of Gross Revenue) increased?

1. 2.3% 2.3.1% 3. 3.8% 4. 5.2%

9. What is the face value per share?
(No.of shares = (Net profit/EPS), Face value = Equity/No.of shares)

1. Rs. 10 2. Rs. 100 3.Rs. 1000 4. Can't say

10. A bonus of 1 share was issued for every 3 held in 2001. In 2002 Gross Receipts = 210 m and Net Margin = 5.3%, what is the EPS in 2002?(Net profit= Net margin as a % of Gross Receipts)

1. Rs. 111 2. Rs. 131 3.Rs. 117 4. Rs. 127

Answer Key & Explanations

Sr. No.	Option	Explanation
1	3	Direct observation; 107.99 – 103.77 = 4.22.
2	4	Direct observation(20)
3	1	Direct observation; 45.36 – 23.34 = 22.02.
4	3	We are asked the total investment. We are investing every year for 4 yrs. So answer should be 200.61 × 4 = 802
5	3	Direct observation (18 years at 10%)
6	2	Growth in revenue: 187.1 – 147.8 = 39.3, which is approx 27% of 147.8
7	2	1.82% of 147.8 = 2.67; 4.65% of 187.1 = 8.7 ; 8.7 – 2.67 = 6 m
8	2	3.9/147.8 × 100 = 2.64 ; 10.8/187.1 × 100 = 5.77 ; Increase in percentage points: 5.77 – 2.64 = 3.13
9	2	From the data you can calculate the number of shares. To do that you need to divide Net profit by EPS = 8,800,000 / 117.33 = 75,000 shares. Face value of the share will be 7,500,000 / 75,000 = Rs. 100.
10	1	Number of new shares issued would be 75,000 / 3 = 25,000. So total number of shares now is 100,000. New net profit is 210 m × 0.053 = 11.1 m. So EPS would be 11.1 m / 0.1 m = Rs. 111.